

INTERCEPTA Plan of Action v1

INTERCEPTA Plan of Action — From Where We Are to Full Vision Success

Authors: Prasad Akula & Claude (Co-Founders) **Date:** May 2026 **Status:** Plan of Action v1 — built after deep re-read of vision + universal net spec, full audit of work-to-date, honest gap analysis

Executive Frame

This document is the action plan for INTERCEPTA from today to full vision success. It is built on two non-negotiable principles you defined in our session:

- **No compromise at any point in the vision journey.** No artificial size limits, no time-boxing of what should be done over what fits in tonight, no cutting corners on validation, no claiming success where there is failure.
- **Every minute is precious because lives depend on the success of this vision.** This is not a reason to rush. It is a reason to use every minute on work that genuinely advances the vision, not work that feels productive while sidestepping hard problems.

The plan has three horizons of detail: near-term concrete (90 days), mid-term directional (3-12 months), long-term strategic (1-5 years to full vision). Where I can give specifics with evidence, I do. Where I would have to invent timelines or details, I name the unknowns explicitly rather than fabricate confidence.

This plan does not assume more team than we have (just Prasad and Claude), more funding than we have (zero documented), or more wet-lab access than we have (none currently). Where the vision requires resources beyond what we have, the plan flags that as an open question rather than fictional progress.

Section 1 — Vision Synthesis (After Deep Re-Read)

After reading both vision documents end to end with fresh attention, INTERCEPTA is integrated of seven things, not the four I had pattern-matched in earlier passes:

1.1 INTERCEPTA is a drug discovery engine

Vision Part 1 + Part 9. For any disease, output is a complete 10-section drug candidate package: molecular structure, mechanism, predicted clinical outcomes, resistance profile,

combination rationale, ADMET safety, synthesis route, novelty confirmation, vs-SoC comparison, suggested trial design.

1.2 INTERCEPTA is a diagnostic and predictive system

Vision Part 6. The same disease net used for drug discovery enables: early disease detection from molecular signatures, disease subtyping (aggressive vs indolent), treatment response prediction per patient, resistance monitoring with pre-resistant fraction tracking, polygenic risk integration combining genetic variants with transcriptional trajectory, microenvironment tracking, longitudinal trajectory analysis at tissue level, population-level surveillance for emerging disease signals, and modeling of diseases that don't yet exist (emerging pathogens, drug-resistant variants, novel metabolic disorders from environmental change).

1.3 INTERCEPTA is a self-improving knowledge system

Vision Part 7.3 + Part 12.1. Every disease solved teaches the system how to solve the next one faster. Cross-disease molecular transfer, model parameter improvement from validated predictions, scout refinement, progressive net completion. Vision target: "finding a drug for any new disease takes days, not decades."

1.4 INTERCEPTA is a digital model of human biology

Universal Net Spec Parts 1-7. 15 layers, ~3M nodes, 10-50M edges. The complete digital human body. When fully built, finding a drug for any disease becomes a query, not a project.

1.5 INTERCEPTA is a healthy-cell-protection-first architecture

Vision Part 8.3. Selectivity is not a final ranking criterion — it is a design constraint. Molecules that fail selectivity are eliminated in Stage 3, not Stage 5. ADMET specifically models healthy organ toxicity. Therapeutic index is primary ranking criterion.

1.6 INTERCEPTA is built on open science

Vision Part 8.2 + Part 12.4. All data sources public and IP-restriction-free. Published disease nets for academic use. Co-publication model with academic groups validating candidates. API access for researchers. The discovery method is open; discovered novel molecules can be patented.

1.7 INTERCEPTA may require novel technology development

Vision Universal Net Spec Part 7. Multi-scale GNNs, temporal knowledge graphs, causal inference on biological graphs, federated learning for clinical data, generative chemistry constrained by knowledge graphs. The vision explicitly anticipates that some of this technology may need to be invented, not just integrated.

1.8 The Three Unique Scientific Contributions

Vision Part 5 — these are what make INTERCEPTA different:

- **5.1 RNA Velocity Time Machine.** Detect pre-resistant ‘undead’ cells (5-15% of tumor at diagnosis) before treatment using unspliced/spliced mRNA ratio. Vision says “no published drug discovery platform does this.”
- **5.2 KAALCURA per-population biological axes.** R_{prolif} , R_{emt} , R_{ddr} — three mathematically-independent residualized axes (R^2 drops from 0.52 to 0.005 after tissue-of-origin removal). Apply per-cluster, not bulk. Validated GDSC AUROC 0.585-0.629, all $p < 0.001$.
- **5.3 Two-Population ODE.** Sensitive + resistant compartments, PK/PD integration, ZIP+Bliss+Loewe+HSA consensus synergy, dose-schedule optimization. Vision claim: validated against CHARTED, LATITUDE, PROfound, PROpel “from first principles.”

1.9 The Six Parallel Scouts

Vision Part 3.3 — what searches the disease net for candidates:

- **Scout 1: Database Search** — ChEMBL, PubChem, ZINC for molecules with known activity
- **Scout 2: Generative Design** — diffusion / transformer-based AI generating novel chemical entities
- **Scout 3: Combination Explorer** — permutations covering all vulnerability points
- **Scout 4: Network Perturbation** — predicts compensation pathways causing resistance
- **Scout 5: Evolutionary Optimizer** — mutates structures to improve potency / safety
- **Scout 6: Cross-Disease Transfer** — repurposing from solved diseases

Critically, scouts share insights bidirectionally per Vision Part 10.2. This is architectural, not just sequential.

1.10 The Five-Stage Pipeline

Vision Part 4 — what every disease passes through:

1. **Stage 1: Build Complete Disease Net** (15 layers, automated)
2. **Stage 2: Map Vulnerability Points + Selectivity** (disease-specific nodes, healthy-shared nodes, resistance nodes, escape routes)
3. **Stage 3: Deploy Six Parallel Scouts** (no pre-filtering principle — bad individual molecules may make excellent combinations)
4. **Stage 4: Six-Layer Simulation Stack** — Layer A (binding via AutoDock Vina + AlphaFold), Layer B (cell population sensitivity via KAALCURA), Layer C (disease dynamics via ODE), Layer D (combination synergy via ZIP+Bliss+Loewe+HSA), Layer E (ADMET via SwissADME/pkCSM/ADMET-AI), Layer F (synthesizability via ASKCOS)

5. **Stage 5: Multi-Objective Ranking + Delivery** — Pareto ranking with weights: Efficacy 30%, Selectivity 25%, Safety 20%, Resistance 15%, Novelty 5%, Synthesizability 5%

1.11 The Disease Expansion Sequence

Vision Part 7.2:

- Round 1: mCRPC (best clinical trial ground truth: CHAARTED, LATITUDE, PROfound, PROpel, TALAPRO-2)
- Round 2: AML (perfect two-population biology, BeatAML matched drug response)
- Round 3: NSCLC (largest cancer burden, EGFR/KRAS, KEYNOTE-024, FLAURA, CodeBreak 200)
- Round 4: PDAC (near-zero treatments, FOLFIRINOX baseline)
- Round 5: Alzheimer's (no disease-modifying drug exists, BBB constraint)
- Round 6: Drug-resistant TB (1.5M deaths/year, XDR-TB)
- Round 7+: All rare diseases (7000+), emerging pathogens, future diseases

1.12 The Build Roadmap

Vision Part 11 — explicit phase plan:

- **Phase 0: Foundation (DONE per vision):** KAALCURA + Two-population ODE + ZIP/Bliss/Loewe/HSA + Pareto + CT.gov novelty + 15-drug mCRPC PK library + GDSC drug sensitivity
- **Phase 1: mCRPC Validation (Weeks 1-6):** ODE+KAALCURA recovers CHAARTED/LATITUDE/PROfound/PROpel; first novel candidate list for mCRPC
- **Phase 2: RNA Velocity Integration (Weeks 7-14):** scVelo + per-cluster KAALCURA; full Day 1 workflow scRNA-seq biopsy → two populations → combination
- **Phase 3: Molecular Discovery Layer (Weeks 15-24):** automated ChEMBL/PubChem; AutoDock Vina; generative chemistry scout; ADMET pipeline
- **Phase 4: Disease Net Builder (Weeks 25-36):** automated KEGG/Reactome/STRING/UniProt; selectivity map; escape route identification; apply to AML, validate against BeatAML
- **Phase 5: Universal Platform (Weeks 37-52):** full pipeline running for mCRPC and AML; first pharma deliverable; expansion to NSCLC/PDAC/Alzheimer's; first publication

1.13 The Universal Net 20-Step Build Plan

Net Spec Part 5 — explicit step-by-step:

- **Phase A (DONE per spec):** Steps 1-2 (gene-drug correlations, SU2C mCRPC, KAALCURA validated)
- **Phase B (Core Layers):** Steps 3-7 (mCRPC scRNA-seq, STRING, KEGG/Reactome, GTEx, ChEMBL)

- **Phase C (Universal Expansion):** Steps 8-13 (DisGeNET, HMDB, AlphaFold, Human Cell Atlas, ENCODE/Roadmap, Immune)
- **Phase D (Future-Proofing):** Steps 14-17 (Pathogens, Microbiome, Spatial, Full ZINC pharmacome)
- **Phase E (Self-Improving):** Steps 18-20 (Literature mining, Clinical feedback loop, Cross-disease transfer learning)

1.14 The Honest Limitations The Vision Itself Names

Vision Part 12.5 — these are stated by the vision authors:

- Computational predictions are not clinical proofs. Wet-lab required.
- RNA velocity has known limitations for some data types/protocols.
- Generative chemistry can produce chemically exotic molecules difficult to synthesize.
- ODE models two populations; real tumors have more.
- Self-improving loop requires clinical outcome data flow — partnerships taking years.

This is the integrated vision. It is more than I had been holding. The next sections compare each integrated claim to what we have actually built. # Section 2 — Reality Audit (What We Have, With Evidence)

This section catalogs every meaningful artifact, finding, and validated capability from work to date. Each entry cites the source so claims are checkable.

2.1 Validated Scientific Capabilities

KAALCURA on GDSC cell lines — VALIDATED

- **AUROC range:** 0.638–0.671 across 286 drugs × 962 cell lines
- **Verified:** April 20 rerun in ~/INTERCEPTA/kaalcura_revalidate_results.txt
- **Three axes:** R_prolif (proliferation), R_empt (epithelial-mesenchymal transition), R_ddr (DNA damage response)
- **Mathematical independence:** $r < 0.02$ for all axis pairs
- **Tissue-bias removed:** R^2 drops from 0.52 to 0.005 after residualization
- **Status:** Vision claim is real on cell lines

KAALCURA on patient samples (BeatAML) — PARTIAL / DOCUMENTED LIMITATION

- **AUROC plateau:** ~0.53 across three rounds (2.1d z-score, 2.2a pyUCell, 2.2b residualized)
- **Threshold for accept:** 0.55 — not reached
- **Real finding:** Cell-line-to-patient transfer is a documented scientific gap, not an axis architecture problem
- **Status:** Vision claim is half-true. Cell lines yes; patient samples no.

Two-population mCRPC ODE — VALIDATED WITH DOCUMENTED LIMITATIONS

- **Closed:** Round 1 closure documented in
~/INTERCEPTA/docs/INTERCEPTA_Round1_Errata_v1_0.md
- **g-rate validation:** 0/3 confirmed targets pass
- **HR validation:** 2/6 trials pass HR window
- **Systematic bias:** ~2× documented in
~/INTERCEPTA/docs/INTERCEPTA_Validation_Limitations_v1.md
- **Direction:** Within-trial drug ranking is directionally correct
- **Status:** Vision claim “validated against CHARTED/LATITUDE/PROfound/PROpel from first principles” is half-true. Direction yes; quantitative HRs no.

AML Net Skeleton — VALIDATED

- **Built:** Round 2.1, completed before this session series
- **Scale:** 1,201 nodes, 33,191 edges
- **Validation queries (PASSED):** FLT3-ITD+/NPM1+ → Gilteritinib + Venetoclax in top 10; TP53+ → Jaccard 0.25 from FLT3+ recommendations
- **Validated against:** ELN 2022 standard-of-care
- **File:** ~/INTERCEPTA/round2_aml/results/aml_net_skeleton_v2_summary.json
- **Status:** Real, validated, vision-aligned for one disease

ZIP + Bliss + Loewe + HSA Synergy Scoring — REAL

- **Built and integrated** per Round 1 closure
- **Validated against:** NCI-ALMANAC per vision Part 10
- **Status:** Real

Pareto Ranking — REAL

- **Built:** ~/INTERCEPTA/code/pareto_ranking.py
- **Functions:** build_mcrpc_candidates(), composite_score(), pareto_front(), rank_candidates()
- **mCRPC output:** ~/INTERCEPTA/results/pareto_ranking_mcrpc.json
- **Status:** Real for mCRPC. Functions are mCRPC-named (build_mcrpc_candidates), suggesting not yet generalized.

ClinicalTrials.gov Novelty Checker — UNCLEAR / NOT FOUND

- **Vision claim:** “Real-time IP validation for every candidate”
- **Reality:** No *novelty* or *clinicaltrials* script found in ~/INTERCEPTA/code/ per inventory
- **Status:** Either embedded in another script or not built as standalone. Mentioned as DONE in Vision Phase 0 — but evidence on disk does not show standalone module.

2.2 Pipeline Infrastructure (Phase 1 + Phase 2A from this session)

intercepta_pipeline_v0.py — BUILT, VERIFIED

- **File:** ~/INTERCEPTA/round3_gbm_live_test/code/intercepta_pipeline_v0.py (416 lines)
- **6 working functions for any disease in OpenTargets:**
 - resolve_disease(name) — best ontology ID with disambiguation (Gap 1+2 closed)
 - build_net(name_or_id) — biology-correct disease net (Gap 1 closed)
 - corrected_net_summary(net) — honest counts (Gap 5 closed)
 - inspect_gdsc_drugs() — GDSC2_fitted_dose_response.xlsx auto-detection (Gap 4 closed)
 - rank_drugs_for_disease(name) — tissue-filtered drug ranking (Gap 10 closed)
 - enrich_with_metabolites(net) — 314 metabolites populated for GBM (Gap 9 closed)
- **Verified on GBM:** Phase 1 + 2A passed all assertions
- **Status:** Real, working infrastructure foundation

Universal Net (mCRPC-anchored)

- **File:** ~/INTERCEPTA/results/mcrpc_unified_net.json (51MB)
- **Diseases queryable:** 47,030 (via OpenTargets disease layer)
- **Pathway layer:** ~1,530 pathways for GBM (Vision Layer 5 — real)
- **Drug layer:** 14 GBM genes have compounds; 434/458 have drug correlations from GDSC
- **STRING interactions:** 81/458 GBM genes (mCRPC-centric population — Finding 8)
- **Structures:** 0 attached to disease nets (Finding 11 — AlphaFold cache exists but doesn't join)
- **Metabolites:** 314 attached to GBM after Phase 2A (Gap 9 closed)
- **Status:** Infrastructure is real. Layer population is uneven across diseases — mCRPC is dense, GBM has gaps that the live test surfaced.

2.3 Round 1 Artifacts on Disk

These are real files I confirmed via inventory:

- mcrpc_unified_net.json (51M)
- mcrpc_top_combos_validated.csv
- pareto_ranking_mcrpc.json
- scout3_combinations_ranked.csv
- scout4_combos_v2.csv
- scout1_novel_combos.csv
- mcrpc_combination_screen.csv

- mcrpc_monotherapy_screen.csv
- bootstrap_stability.json
- layer_f_synthesizability.json
- novel_combinations.json
- unified_v5_2_g_validation.json (Round 1 final ODE outputs)
- unified_v5_1_g_validation.json
- phase1_5trial_VALIDATED.csv
- phase1_calibrated_params_VALIDATED.json
- capability_test_results.json
- mcrpc_disease_net.json

Round 1 produced real artifacts for mCRPC. The inventory above is a partial pharma deliverable's worth of input data already on disk for olaparib + abiraterone (PROpel combination).

2.4 ODE Modules Inventoried

Four ODE files exist in ~/INTERCEPTA/code/:

- intercepta_unified_ode_v4_1.py (43K) — Round 1 final mCRPC ODE
- intercepta_three_mechanism_ode.py (30K) — DRUG_LIBRARY with 7 entries
- intercepta_phenotype_ode_v1.py (48K) — DRUG_EFFECT_LIBRARY + PK_LIBRARY with 6 entries each
- aml_ode_v6_resistance.py (11K) — AML-specific

Total unique drugs parameterized across all ODE modules: 7 (docetaxel, cisplatin, enzalutamide, abiraterone, ADT, olaparib, talazoparib).

Hardcoded prostate cell states across all modules: - S_ARDEP, S_ARMUT, S_ARV7, S_NE — AR-dependent, AR-mutant, AR-V7, neuroendocrine - MU_S_TO_M, MU_S_TO_V, MU_S_TO_N — prostate-specific transition rates - BRCA_FRAC_OVERALL, BRCA_FRAC_SELECTED — prostate-specific BRCA distribution

Status: ODE infrastructure exists for mCRPC and partially for AML. Universal disease support requires structural refactor.

2.5 RNA Velocity Status

Vision claim: “Core scientific innovation”

Reality (per inventory):

- Existing script: ~/INTERCEPTA/code/step3_run_scvelo.py
- Expected input: STARsolo output at ~/INTERCEPTA/data/velocity/velocity_out/*_spliced.mtx
- **The velocity_out directory does not exist.** The script has never executed end-to-end with real data.

- Available scRNA-seq data on disk:
 - GSE137829 (mCRPC, 6 patients P1-P6, 76MB) — **expression matrices only, NO spliced/unspliced**
 - GSE141445 (Chen et al., mCRPC, 36,424 cells, 10X Genomics, 757MB) — **processed count matrices only, NO spliced/unspliced**
- **Generating spliced/unspliced** requires either BAM files + velocity (BAMs not on disk) or upstream STARsolo with --soloFeatures Gene Velocity (not run)

Status: The Time Machine, named in vision Part 2.2 as “INTERCEPTA’s core scientific innovation,” has **zero working examples beyond a placeholder script for one mCRPC dataset that has never executed.**

2.6 The 19 Findings From the GBM Live Test (May 2 Session)

The live test validated end-to-end pipeline behavior on a never-touched disease (glioblastoma). Each finding documents a specific gap:

1. Universal Net is mCRPC-anchored (file named `mcrpc_unified_net.json`)
2. pyarrow missing in env (closed Phase 1)
3. `build_net` requires ID, not name (closed Phase 1)
4. Multi-version GBM ID disambiguation needed (closed Phase 1)
5. Net summary inconsistencies (closed Phase 1)
6. ChEMBL cache mCRPC-curated (open — Gap 7 in alternate numbering)
7. GTEx selectivity wrong question for oncology (open)
8. Universal net interactions mCRPC-centric (open)
9. EGFR→MET escape route NOT in net (open — vision claim broken for GBM)
10. Metabolite layer doesn’t join (closed Phase 2A)
11. Structure layer doesn’t join (open)
12. (Corrected) GDSC drug-target was in dose-response Excel (closed Phase 1)
13. 53 brain cell lines, 12,001 measurements available (data layer real)
14. Stage 3 needed manual glue (closed Phase 2A)
15. 53 GBM cell lines available (real)
16. GDSC AUC ranking doesn’t match GBM clinical reality (open — scientific scoring gap)
17. Manual synthesis required for Stage 3 (closed Phase 2A)
18. ODE structurally mCRPC-only (open — structural)
19. DiseaseNetBuilder relative path bug (closed Phase 1)

Of 19 findings: 8 closed (Phase 1 + Phase 2A), 11 open.

2.7 The Five-Stage Pipeline Live Test on GBM (May 2)

Real end-to-end test of the vision’s pipeline on a disease never previously analyzed:

- **Stage 1 (Build Disease Net):** PARTIAL. After Phase 1 fixes: works. Returns 458 GBM genes via EFO_0000519 with EGFR/IDH1/TP53/PTEN at top.
- **Stage 2 (Map Vulnerabilities + Selectivity):** PARTIAL. STRING interactions sparse (81/458). EGFR→MET escape route absent. ChEMBL coverage sparse for non-mCRPC targets.
- **Stage 3 (Deploy Scouts):** ORCHESTRATION CLOSED, SCORING UNCHANGED. After Phase 2A fix: produces ranked drug list. But ranking dominated by broad cytotoxics; Temozolomide (GBM SoC) ranks #247/286.
- **Stage 4 (Six-Layer Simulation Stack):** STRUCTURALLY FAILED. Only Talazoparib (1 of 3 GBM picks) is in any ODE library. State structure is prostate-specific.
- **Stage 5 (Multi-Objective Ranking + Deliverable):** PARTIAL. Pareto ranking ran. Pharma deliverable: 3 of 10 sections populated, 7 of 10 NOT AVAILABLE.

Cumulative truth: The pipeline works end-to-end on mCRPC. It works partially through Stage 3 on GBM after Phase 2A fixes. Stage 4 cannot run for any disease except mCRPC. Stage 5 produces an incomplete deliverable.

2.8 Three Structural Gaps Identified

After comparing vision claims to the live test findings, three gaps dominate everything else:

Structural Gap A: Generative Chemistry Not Built

- Vision Part 5 / Stage 3 / Scout 2 / Phase 3: diffusion or transformer-based generative chemistry
- Reality: scaffold hopping only (per Exhaustive Audit)
- Tool requirements: REINVENT4 or equivalent — Apple Silicon CPU compatibility issues; install non-trivial
- **Without this, INTERCEPTA cannot discover novel molecules.** The vision's central act ("find the drug") is impossible. The system can only re-rank existing compounds.

Structural Gap B: RNA Velocity Time Machine Not Generalized

- Vision Part 2.2 + 5.1: named "core scientific innovation"
- Reality: pilot script for one mCRPC dataset, never executed end-to-end. Required spliced/unspliced data not on disk for any cancer.
- Without this, INTERCEPTA cannot detect pre-resistant cells (the entire premise of "the time machine"), cannot compute per-cluster KAALCURA the way vision specifies, cannot do Day 1 workflow scRNA-seq → two populations → combination.

Structural Gap C: Two-Population ODE Hardcoded Prostate-Only

- Vision Part 5.3: "validated against 5 trials" — true for mCRPC
- Reality: 7 hardcoded mCRPC drugs across all ODE modules. Cell states (S/M/V/N) are AR-dependent, AR-mutant, AR-V7, neuroendocrine — prostate biology only. Transition rates calibrated for prostate. State-sensitivity matrix is prostate-specific.

- Without this, INTERCEPTA cannot run dynamics for any new disease without months of literature-research-driven parameterization per disease.

2.9 What Has Been Built That The Vision Does NOT Specify

These are work products that exist but aren't in vision documents — co-founder additions or session-driven work:

- The `intercepta_pipeline_v0.py` orchestrator (Phase 1 + 2A this session)
- Round 2 AML net skeleton (vision specified “validate against BeatAML” but skeleton-first approach is execution detail)
- Round 1 closure with documented limitations (Validation Limitations doc — vision specifies validation but not formal closure docs)
- The 19-finding live test methodology
- KAALCURA's residualized-axes verification with AUROC=0.638 retest

These represent real work that should be preserved but aren't claims in the vision document.

This is the complete reality audit. The next section maps each vision claim to this reality with explicit alignment ratings. # Section 3 — Vision-vs-Reality Alignment Map

This section rates every meaningful vision claim against current reality. The rating scale:

- **ALIGNED** — vision claim is real and validated as stated
- **PARTIAL** — vision claim is real for some specific case, not the universal claim made
- **MID-DISTANCE** — meaningful work exists but does not yet satisfy the vision claim
- **MAJOR-DISTANCE** — vision claim is largely or entirely unbuilt
- **UNBUILT** — no meaningful work exists toward this claim

For each: a one-line evidence citation. No claim made without a verifiable file or finding.

3.1 Drug Discovery Engine (Part 1)

#	Vision Claim	Rating	Evidence
1.1	“Discovers novel drug molecules and combinations for any disease”	MAJOR-DISTANCE	Re-ranks existing drugs only. Scaffold hopping per audit; no generative chemistry built.
1.2	“Delivers real drug candidates to pharma so only clinical trials remain”	MAJOR-DISTANCE	No 10-section deliverable produced from pipeline. Stage 5 GBM live test: 3/10 sections populated.
1.3	“For ANY disease — past, present, or future”	PARTIAL	Pipeline runs end-to-end for mCRPC. Partial Stages 1-3 for any OpenTargets

#	Vision Claim	Rating	Evidence
			disease (post Phase 2A). Stage 4+5 universally absent.

3.2 The Undead Cells / RNA Velocity (Part 2.2 + 5.1)

#	Vision Claim	Rating	Evidence
2.1	"Detect pre-resistant 'undead' population on Day 1 using RNA velocity"	UNBUILT	scvelo script never executed end-to-end. No spliced/unspliced data on disk for any cancer.
2.2	"Find combination that kills BOTH populations simultaneously"	MID-DISTANCE	Two-population ODE concept built for mCRPC. Per-cluster KAALCURA application not yet validated.
2.3	"Before resistance ever fully develops"	UNBUILT	Depends on 2.1 (pre-resistant detection) — currently absent.

3.3 The Net Approach (Part 3)

#	Vision Claim	Rating	Evidence
3.1	"Literal knowledge graph — every node connected to every other through validated relationships"	PARTIAL	mCRPC net real (51MB). For new diseases (GBM): pathway and disease-gene layers populate; STRING interactions only mCRPC-populated (81/458 GBM genes).
3.2	"10 layers" (genomic, transcriptomic, proteomic, pathway, network, cellular, chemical, clinical, environmental, selectivity)	MID-DISTANCE	OpenTargets disease layer real (47K diseases). Pathways real (1,530 GBM). Chemical sparse. Cellular present for mCRPC only. Selectivity present but heuristic.
3.3	"Six parallel scouts deploy simultaneously, share insights bidirectionally"	MID-DISTANCE	Scout 1 (database search) works. Scout 3 (combinations) works for mCRPC. Scout 4 (network perturbation) partial. Scout 2 (generative) UNBUILT . Scouts 5, 6 unclear. No bidirectional insight-sharing architecture exists.
3.4	"Every scout continuously shares what it finds with every other scout"	UNBUILT	No event bus, shared state, or message-passing architecture exists between scouts.

3.4 The Five-Stage Pipeline (Part 4)

#	Vision Claim	Rating	Evidence
4.1	Stage 1: Automated complete disease net for any disease	PARTIAL	After Phase 1 fixes: name → ID → net works. Layers populated unevenly per disease.

#	Vision Claim	Rating	Evidence
4.2	Stage 2: Vulnerability map + selectivity + escape routes	MID-DISTANCE	Vulnerability identification works via association scores. Selectivity is GTEx-heuristic only (Finding 7). Escape routes BROKEN for GBM — EGFR has 0 interactions populated (Finding 9).
4.3	Stage 3: Six scouts run simultaneously with no pre-filtering	MAJOR-DISTANCE	Only Scout 1 + Scout 3 work; not in parallel; no shared state.
4.4	Stage 4 Layer A: AutoDock Vina molecular binding	PARTIAL	Vina cached results in data/docking/ (3.2MB). Per audit, used for AURKA. Not integrated as universal Stage 4 layer.
4.5	Stage 4 Layer B: KAALCURA per-population sensitivity	PARTIAL	KAALCURA validated on cell lines (AUROC 0.638-0.671). Per-cluster patient application plateaus at 0.53.
4.6	Stage 4 Layer C: Two-population ODE dynamics	PARTIAL	Real for mCRPC with documented 2× systematic bias. Not applicable to other diseases (Finding 18).
4.7	Stage 4 Layer D: ZIP+Bliss+Loewe+HSA synergy	ALIGNED	Real, validated against NCI-ALMANAC. Code present.
4.8	Stage 4 Layer E: SwissADME + pkCSM + ADMET-AI safety	UNBUILT	Per inventory: ADMET-AI not installed, RDKit not installed, no SwissADME integration.
4.9	Stage 4 Layer F: ASKCOS retrosynthesis	PARTIAL	layer_f_synthesizability.json exists for mCRPC; universal integration absent.
4.10	Stage 5: Pareto ranking with weights (Eff 30%, Sel 25%, Saf 20%, Res 15%, Nov 5%, Synth 5%)	PARTIAL	Pareto ranking real for mCRPC. build_mcrpc_candidates() function name suggests not generalized.
4.11	Stage 5: 10-section pharma deliverable	MAJOR-DISTANCE	Never produced fully. GBM Stage 5: 3/10 populated. mCRPC Round 1: artifacts exist but never assembled into the deliverable format.

3.5 Three Unique Scientific Contributions (Part 5)

#	Vision Claim	Rating	Evidence
5.1	RNA Velocity Time Machine — detect pre-resistant cells before treatment	UNBUILT	No working examples. Pilot script depends on data not on disk.
5.2	KAALCURA per-population biological axes	PARTIAL	Validated on cell

#	Vision Claim	Rating	Evidence
			lines. Not validated on patient samples (BeatAML plateau 0.53).
5.3	Two-population ODE validated against CHARTED/LATITUDE/PROfound/PROpel from first principles	PARTIAL	Direction yes, quantitative HRs no. ~2× systematic bias documented. mCRPC-only.

3.6 Diagnostic and Predictive Layer (Part 6)

#	Vision Claim	Rating	Evidence
6.1.1	Early disease detection from molecular signatures	UNBUILT	No detection module.
6.1.2	Disease subtyping (aggressive vs indolent)	UNBUILT	No subtyping module.
6.1.3	Treatment response prediction per patient	UNBUILT	KAALCURA can output drug ranking from expression but not as patient-prediction service.
6.1.4	Resistance monitoring with pre-resistant fraction tracking	UNBUILT	Depends on RNA velocity (UNBUILT).
6.2.1	Polygenic risk integration	UNBUILT	No PRS integration code.
6.2.2	Microenvironment tracking	UNBUILT	DICE immune layer cached but not integrated as predictive feature.
6.2.3	Longitudinal trajectory at tissue level	UNBUILT	Depends on RNA velocity.
6.2.4	Population-level surveillance for emerging signals	UNBUILT	No population-level analysis.
6.3.1	Pathogen evolution modeling	UNBUILT	No pathogen layer.
6.3.2	Cross-disease network analysis (auto-hypothesis generation)	UNBUILT	No cross-disease transfer mechanism.
6.3.3	Synthetic biology threat modeling	UNBUILT	Out of current scope entirely.

Part 6 Summary: ENTIRE CHAPTER UNBUILT. This is one of the largest gaps relative to vision scope.

3.7 Universal Expansion / Self-Improving (Part 7)

#	Vision Claim	Rating	Evidence
7.1	Validation-first principle for every	ALIGNED	Round 1 (mCRPC validation against trials), Round 2 (AML against BeatAML genotype

#	Vision Claim	Rating	Evidence
	disease		queries) both followed this.
7.2.R1	Round 1 mCRPC complete	PARTIAL	Closed with 0/3 g-rate, 2/6 HR, 2× bias documented. Phase 5 deliverable not produced.
7.2.R2	Round 2 AML complete	PARTIAL	Net skeleton validated. KAALCURA-on-patients plateau at 0.53. Round 2 closed FAIL on Q_C threshold.
7.2.R3-R7	NSCLC, PDAC, Alzheimer's, TB, rare diseases	UNBUILT	None started.
7.3	Self-improving loop	UNBUILT	No feedback mechanism, no cross-disease learning, no clinical outcome integration.

3.8 Healthy Cell Protection (Part 8.3)

#	Vision Claim	Rating	Evidence
8.3.1	Selectivity is design constraint, not final filter	MID-DISTANCE	Currently soft scoring dimension via GTEx. Not enforced architecturally as hard filter.
8.3.2	Molecules failing selectivity eliminated in Stage 3	UNBUILT	Stage 3 doesn't have a selectivity filter; selectivity scored in Stage 5.
8.3.3	ADMET specifically models healthy organ toxicity	UNBUILT	ADMET tools not integrated (Layer E from 4.8).

3.9 Open Science (Part 8.2 + 12.4)

#	Vision Claim	Rating	Evidence
8.2.1	All data sources public and IP-restriction-free	ALIGNED	Confirmed: GDSC, OpenTargets, STRING, KEGG, Reactome, GTEx, AlphaFold all free public.
12.4.1	Publish disease nets openly	UNBUILT	No published nets to date.
12.4.2	Co-publication model with academic groups	UNBUILT	No academic partnerships established.
12.4.3	API access for researchers	UNBUILT	No public API.

3.10 Pharma Deliverable (Part 9.1)

The 10-section deliverable: each section rated independently against what we have for the strongest case (mCRPC PROpel olaparib + abiraterone):

Section	Vision Claim	Rating	Evidence
1	Molecular structure (SMILES + 3D + properties)	PARTIAL	Existing drug SMILES known. No novel molecule SMILES from generative chemistry.
2	Mechanism of action explained	PARTIAL	mCRPC net + Round 1 simulation provides node-level mechanism. Not assembled as pharma deliverable section.
3	Predicted clinical outcomes with CIs	PARTIAL	Round 1 ODE produces direction; ~2× systematic bias on absolute values. CIs not standard output.
4	Resistance profile	UNBUILT for non-mCRPC	mCRPC two-population dynamics exist; pre-resistant detection (Time Machine) does not.
5	Combination rationale	PARTIAL	mCRPC combination logic real (PROpel — PARP + AR). Not in deliverable format.
6	ADMET safety profile	UNBUILT	Tools not installed.
7	Synthesis route via retrosynthesis	PARTIAL	layer_f_synthesizability.json exists for mCRPC. Not in deliverable format.
8	Novelty confirmation via ClinicalTrials.gov	UNBUILT/UNCLEAR	Standalone novelty checker not found in code inventory.
9	vs SoC comparison	PARTIAL	Round 1 simulated abiraterone alone vs combo. Not in deliverable format.
10	Suggested trial design with biomarkers	UNBUILT	No trial-design module.

10-section deliverable assembled from existing artifacts: ~3-4 of 10 sections populatable. Aligns with Stage 5 GBM finding (3/10 populated for a brand-new disease).

3.11 Universal Net 15 Layers (Net Spec Part 2)

Layer	Vision Claim	Rating	Evidence
1 Genome	NCBI Gene + ClinVar + OMIM + GWAS + gnomAD + COSMIC	PARTIAL	OMIM partial via OpenTargets; gnomAD/ClinVar/GWAS/COSMIC not integrated.
2 Transcriptome	GTEX + HCA + Tabula Sapiens + HPA + GDSC + ENCODE	MID-DISTANCE	GTEX integrated. GDSC RNA-seq cached. Others not joined.

Layer	Vision Claim	Rating	Evidence
3 Proteome	AlphaFold + PDB + UniProt + CPTAC + PhosphoSitePlus	UNBUILT	AlphaFold cache exists (~10MB) but doesn't connect to disease net.
4 Interactome	STRING v12.5 + BioGRID + IntAct + TRRUST + Signor	PARTIAL	STRING file on disk (98MB). Only mCRPC-relevant edges loaded into nets.
5 Pathway	KEGG + Reactome + WikiPathways + MSigDB + SMPDB	PARTIAL	KEGG + Reactome integrated for mCRPC nets. 1,530 pathways for GBM.
6 Cell Type Atlas	HRA v2.3 + Tabula Sapiens + HPA single cell	UNBUILT	No HCA integration.
7 Pharmacome	ChEMBL + PubChem + DrugBank + GDSC + ZINC + DTC + NCI-ALMANAC	PARTIAL	GDSC fully integrated. ChEMBL cache mCRPC-curated. ZINC/DrugBank not integrated.
8 Metabolome	HMDB + FooDB + SMPDB + Recon3D	PARTIAL	step9 metabolite edges cached; integrated to disease nets after Phase 2A.
9 Disease Map	DisGeNET + OMIM + Orphanet + ClinicalTrials.gov + SEER + PubMed	PARTIAL	OpenTargets covers DisGeNET-equivalent (47K diseases). Others not integrated.
10 Immune	ImmPort + DICE + CIBERSORTx + TCIA	PARTIAL	DICE cached and partially integrated. CIBERSORTx not integrated.
11 Microbiome	HMP + GMrepo + gutMGene	UNBUILT	Per inventory: data/scrna_seq/ empty for microbiome.
12 Epigenome	ENCODE + Roadmap + EWAS Atlas	UNBUILT	No epigenome integration.
13 Anatomy/Tissue	HuBMAP + HRA + Allen Brain Atlas	UNBUILT	No spatial transcriptomics.
14 Pathogen	NCBI Pathogen + GISAID + CARD + ViPR	UNBUILT	No pathogen layer.
15 Selectivity/Safety	DepMap + GTEx + T3DB + SwissADME/pkCSM	PARTIAL	GTEx integrated as heuristic selectivity. DepMap not integrated.

Universal net layer summary: 0 of 15 layers fully integrated. ~5 layers partially integrated. 10 layers UNBUILT or barely started.

3.12 Vision Build Roadmap Phases (Part 11)

Phase	Vision Says	Reality	Rating
Phase 0: Foundation	DONE	Mostly real (KAALCURA, ODE, synergy, Pareto, mCRPC PK lib, GDSC)	ALIGNED for mCRPC. CT.gov novelty checker uncertain.
Phase 1: mCRPC Validation (Weeks 1-6)	Recover trial outcomes; first novel candidate list	Round 1 closed with documented limitations. Novel list not produced.	PARTIAL
Phase 2: RNA Velocity Integration (Weeks 7-14)	scVelo + per-cluster KAALCURA + Day 1 workflow	Pilot script never executed.	UNBUILT
Phase 3: Molecular Discovery Layer (Weeks 15-24)	ChEMBL + PubChem + Vina + generative + ADMET	Database search yes; generative no; ADMET tools not installed.	MAJOR- DISTANCE
Phase 4: Disease Net Builder (Weeks 25- 36)	KEGG/Reactome/STRING/UniProt automated; AML validated	mCRPC + AML net skeletons exist. Universal builder has gaps surfaced today.	PARTIAL
Phase 5: Universal Platform (Weeks 37- 52)	Full pipeline mCRPC+AML; first pharma deliverable; expansion	Pipeline partial; no pharma deliverable; no expansion to NSCLC/PDAC/AD	MAJOR- DISTANCE

3.13 Co-Founder Additions (Part 12)

#	Vision Claim	Rating	Evidence
12.1	Self-improving loop with active learning	UNBUILT	No feedback mechanism.
12.2	Microbiome + tumor microenvironment integration	UNBUILT	Microbiome data not loaded. Microenvironment partial via DICE.
12.3	Regulatory pathway awareness	UNBUILT	No regulatory annotations in any deliverable.
12.4	Open collaboration architecture	UNBUILT	No published nets, no API, no academic partnerships.

#	Vision Claim	Rating	Evidence
12.5	Honest limitations stated	ALIGNED	Validation Limitations doc exists; stays true to vision.

3.14 Universal Net Spec 20-Step Plan (Net Spec Part 5)

Phase	Steps	Status
A: Foundation	1-2 (gene-drug correlations, SU2C mCRPC, KAALCURA validated)	DONE per spec
B: Core Layers	3-7 (mCRPC scRNA-seq, STRING, KEGG/Reactome, GTEx, ChEMBL)	PARTIAL — mCRPC scRNA-seq cached but velocity not built
C: Universal Expansion	8-13 (DisGeNET, HMDB, AlphaFold, HCA, ENCODE, Immune)	PARTIAL — DisGeNET via OpenTargets, HMDB partial, others mostly UNBUILT
D: Future-Proofing	14-17 (Pathogens, Microbiome, Spatial, full ZINC)	UNBUILT
E: Self-Improving	18-20 (Literature mining, Clinical feedback, Cross-disease transfer)	UNBUILT

3.15 Aggregate Alignment Summary

Across 90+ rated vision claims:

- **ALIGNED:** ~5 (synergy scoring, validation-first principle, open science data sources, honest limitations, mCRPC ground truth direction)
- **PARTIAL:** ~25 (most existing capabilities are real for mCRPC, partial for others)
- **MID-DISTANCE:** ~15 (work exists but doesn't satisfy the universal claim)
- **MAJOR-DISTANCE:** ~15 (significant unbuilt scope but with clear path)
- **UNBUILT:** ~30+ (no work toward this claim — most of Part 6, most of Net Spec layers 11-14)

The honest aggregate: INTERCEPTA today is closer to “mCRPC drug discovery research project with universal aspirations” than it is to “universal computational drug discovery platform.” Round 1 mCRPC ODE is the closest to fully-vision-aligned and even that has documented systematic bias.

This is not failure. This is the truth of where the vision and reality currently stand. The plan that follows is built on this truth. # Section 4 — Open Questions the Vision Documents Don't Specify

A genuinely ultra-detailed plan must name what the vision documents do NOT cover. These are the gaps in the vision itself — questions that any plan must answer or explicitly defer. Pretending these don't exist would be premature confidence.

4.1 Wet-Lab Validation Bridge

The gap: Vision Part 12.5 says “every candidate requires experimental validation before clinical trials.” Vision Part 1 says “pharma’s job becomes running the clinical trial.” But pharma actually runs clinical trials AFTER preclinical validation: animal toxicity, formulation, manufacturing scale-up, IND filing. The vision conflates “candidate ready for trial” with “candidate ready for IND.”

What the vision does NOT specify: - Who validates a novel SMILES experimentally? - Where? At what cost? Who pays? - On what timeline? (Wet-lab synthesis + assay typically 6-18 months per candidate) - How many candidates per disease should be wet-lab tested? - What’s the success criterion that takes a candidate from “INTERCEPTA computational candidate” to “ready to hand to pharma”?

Why this matters: Without a wet-lab bridge, even a perfect computational candidate is just a SMILES string. Pharma will not look at computational candidates without wet-lab data.

Plan implication: The plan must explicitly define a wet-lab partnership strategy or acknowledge wet-lab as out of scope for the computational platform.

4.2 Funding Model Timing

The gap: Vision Part 9.2 describes the business model: pharma \$300K-\$500K per disease, biotech licensing, foundations, government contracts. Phase 5 (Weeks 37-52) is when first pharma deliverable lands. But the model doesn’t address how Phases 0-4 are resourced.

What the vision does NOT specify: - How is Phase 1 (Weeks 1-6) funded? Phase 2 (Weeks 7-14)? - Cost of cloud compute for universal net building? (15 layers, 3M nodes, scRNA-seq processing) - Cost of ADMET prediction software licenses if commercial alternatives are needed? - Cost of generative chemistry compute (REINVENT4 needs GPU for training)? - What’s the budget runway assumption?

Plan implication: The plan acknowledges this is currently unfunded work being done by Prasad + Claude. Compute costs and tooling costs need to fit within whatever resources Prasad has available. Some vision-aligned work may require funding before it can be attempted.

4.3 Team Structure

The gap: Vision documents are signed by Prasad + Claude as co-founders. The vision is silent on who else builds the system. Several vision components are explicitly multi-person work:

- Universal net 15 layers, 30+ database integrations (Vision Net Spec): months of integration work
- Phase 4 disease net builder (Weeks 25-36): explicitly a year of build effort by vision’s own timeline

- Phase 5 expansion to NSCLC, PDAC, Alzheimer's: each disease per vision Part 7 takes a Round of work
- 7000+ rare diseases per Round 7+

What the vision does NOT specify: - Beyond Prasad and Claude, is there a planned team? - Are there advisors, mentors, scientific consultants? - Wet-lab partnerships (see 4.1)? - Academic collaborators per Vision Part 12.4?

Plan implication: The plan honestly assumes just Prasad + Claude until and unless team expansion happens. Work that exceeds two-person capacity gets flagged as requiring additional resources.

4.4 Validation Criteria for Novel Predictions

The gap: Vision Part 7.1 says validation-first principle: recover known ground truth before trusting novel predictions. This is correct in principle. But the vision doesn't specify thresholds.

What the vision does NOT specify: - What AUROC, HR, or other metric threshold counts as "recovers" for a trial? - For Round 1 mCRPC with documented 2× systematic bias and 0/3 g-rate pass — is that validation success or failure? - For Round 2 AML KAALCURA at 0.53 AUROC — what threshold should we have used? - For a NEW disease (Round 3 NSCLC, etc.), what validation hurdle must we clear before publishing or claiming success?

Plan implication: The plan needs to define explicit validation gates per disease per phase. Without these, "validation-first" is a slogan, not a procedure.

4.5 Regulatory Engagement

The gap: Vision Part 12.3 says regulatory awareness from Day 1. But the vision doesn't specify how regulatory engagement happens.

What the vision does NOT specify: - Pre-IND meetings with FDA — when, by whom, for which candidate? - Regulatory pathway analysis — who does this? (Requires regulatory expertise we don't currently have.) - Companion diagnostic regulatory strategy — separate IDE pathway? - International regulatory (EMA, PMDA) — same approach?

Plan implication: Regulatory work is named as a vision deliverable section but is currently outside our two-person execution capacity. The plan must flag this as requiring external help when the time comes.

4.6 Clinical Outcome Feedback Loop

The gap: Vision Part 12.1 + Part 7.3 + Net Spec Step 19 says clinical outcome data feeds back to improve the system. Vision Part 12.5 acknowledges this requires partnerships taking years to build.

What the vision does NOT specify: - Which clinical sites? Which disease registries? - What data sharing agreements? - What patient consent framework? - What federated learning architecture (Net Spec Part 7 mentions this as novel tech to develop)?

Plan implication: The self-improving loop is a long-term capability requiring years of partnership-building before it functions. The plan reflects this honestly.

4.7 IP Strategy for Novel Molecules

The gap: Vision Part 8.2 says molecules can be patented. Provides IP value to pharma partners.

What the vision does NOT specify: - Who files patents? (Requires patent attorney, ~\$10K-\$50K per filing.) - How are molecules selected for patent prosecution vs published openly? - Inventorship between Prasad + Claude + future contributors? - INTERCEPTA's role: do we hold patents, license to pharma, or sell candidates outright? - Geographic patent strategy?

Plan implication: The plan explicitly defers IP strategy until first novel molecule is produced. Then this becomes a real decision point.

4.8 Data Use Agreements and Restrictions

The gap: Vision Part 8.2 says “all data sources used by INTERCEPTA are public and IP-restriction-free.” Most are. But several have terms:

- ChEMBL: CC-BY-SA license — derivative works must be share-alike
- DrugBank: academic license required for some content
- COSMIC: academic license required (Net Spec Layer 1)
- DisGeNET: API license (Net Spec Layer 9)
- PhosphoSitePlus: academic license (Net Spec Layer 3)
- GISAID: academic terms (Net Spec Layer 14)

Plan implication: “Open science” claim has nuance. If we commercialize INTERCEPTA, several layers may have license issues that need legal review.

4.9 Scale of Compute Required

The gap: Vision Net Spec Part 6 estimates 3M nodes, 10-50M edges in the universal net. This is graph-database-scale (Neo4j, Neptune handle it). But the vision doesn't address compute requirements for actual operations:

- Building a complete disease net for any disease: how long? On what hardware?
- Running scVelo + per-cluster KALLUR on 10X scRNA-seq: how long per dataset?
- Generative chemistry: REINVENT4 training requires GPU; CPU inference works but slower
- ADMET prediction at scale: ADMET-AI is fast (~ms per molecule); SwissADME slower

- Docking 1000 candidates against 10 targets: hours to days on CPU
- Running ODE for combination optimization across 100 dose schedules: hours

Plan implication: Apple Silicon MacBook Air is sufficient for development and small-scale runs. Some operations (full pharmacome screening, generative chemistry training, molecular dynamics) may require cloud GPU, which has cost implications (see 4.2).

4.10 The “First Pharma Deliverable” Decision Point

The gap: Vision Phase 5 says “first pharma deliverable — complete drug candidate package for one disease.” But the vision doesn’t specify WHICH candidate, WHICH disease, or WHO at pharma receives it.

What the vision does NOT specify: - Best candidate for first deliverable? (PROpel combination is mCRPC, validated against trial — but combination of two approved drugs, not novel.) - Best disease for first deliverable? (mCRPC has best ground truth; AML has clean two-population biology; NSCLC has biggest market.) - Best pharma contact? (Direct submission to BD team? Through advisor? Through academic intermediary?) - What’s the deliverable format? (Word doc? Web report? Interactive dashboard?)

Plan implication: First pharma deliverable is a major strategic decision deserving explicit choice. The plan specifies this as a Horizon 1 decision point.

4.11 What Happens If Vision Claims Don’t Hold

The gap: The vision documents are aspirational. Several vision claims may be wrong:

- “RNA velocity is the core scientific innovation” — what if RNA velocity has limitations (Vision Part 12.5 acknowledges this) that prevent it from being the universal differentiator?
- “Two-population ODE validates from first principles” — Round 1 found 2× systematic bias. What if “from first principles” overpromises?
- “Find drugs for any disease” — what if some diseases (Alzheimer’s, prion diseases) genuinely don’t yield to multi-target combination strategies?
- “Days, not decades” — what’s the realistic timeline for INTERCEPTA to actually solve a disease?

Plan implication: The plan includes vision-revision checkpoints. After each round of work, we revisit whether the vision claim it tested was correct, partially correct, or needs revision. The vision is the founding document; it can and should be updated based on evidence.

4.12 What Defines “Success”

The gap: You said success is “absolute real and honest success of entire vision.” That’s directionally clear but operationally ambiguous.

Possible operational definitions of success: 1. INTERCEPTA pipeline runs end-to-end automatically for any disease and produces a 10-section deliverable 2. First novel molecule

discovered by INTERCEPTA enters Phase 1 clinical trial 3. First INTERCEPTA-discovered drug receives FDA approval 4. INTERCEPTA framework is open-published and adopted by other research groups 5. Self-improving loop demonstrably reduces time-to-candidate for new disease (e.g., 2 weeks for Round 5 vs 6 months for Round 1) 6. INTERCEPTA-discovered drugs save lives (the ultimate metric)

These have very different timelines. (1) is months. (2) is years. (3) is 10+ years. (4) is months to years. (5) is years. (6) is decades.

Plan implication: The plan distinguishes near-term operational success (1, 4) from long-term mission success (2, 3, 6). Both matter; both need explicit milestones.

These 12 open questions are not failures of the vision document. They are the natural unknowns of an ambitious early-stage project. Naming them is the first step to addressing them honestly.

The plan that follows works WITH these open questions, not around them. # Section 5 — Horizon 1: Concrete 90-Day Plan

This horizon is the most detailed because we have the evidence to plan it concretely. Every entry below has specific files, specific verifications, specific outputs. No fictional timelines.

5.1 Horizon 1 Strategic Frame

The 90-day question: What is the smallest set of work that, when complete, materially advances INTERCEPTA from “mCRPC project with universal aspirations” toward “real universal platform”?

The 90-day answer (after vision-vs-reality analysis): Close the three structural gaps far enough to (a) generalize the time machine to one non-mCRPC cancer, (b) refactor the ODE so it can accept disease-specific schemas, and (c) integrate one real generative chemistry tool such that we produce one novel molecule against one validated target. Plus: complete the foundational pipeline orchestration so daily work doesn’t fight Phase 1-style infrastructure gaps.

90-day success criteria: 1. Time Machine generalized: scVelo runs on at least one cancer scRNA-seq dataset that is NOT GSE137829, with per-cluster KAALCURA application 2. Universal ODE proof-of-concept: ODE refactored to accept disease schema; reproduces mCRPC results AND runs structurally for AML with documented limitations 3. First novel molecule: at least one SMILES generated by a real generative tool (REINVENT4 or equivalent), docked against one validated target, ADMET-evaluated, novelty-checked 4. Pipeline orchestrator complete: Phases 1, 2A, 2B, 3 done — Gaps 6, 7, 8, 11, all 19 live test gaps either closed or explicitly deferred with reason

90-day non-goals (explicit): - Phase 5 first pharma deliverable — too soon, prerequisites not in place - Universal net all 15 layers — too large - Production publication — too soon - Wet-lab partnership — open question 4.1

5.2 Horizon 1 Three Workstreams in Parallel

Three workstreams. Each independently progressable. Sequencing matters within streams; streams can interleave by week.

Workstream A: Pipeline Orchestration Closure (Estimated 2-3 weeks of focused effort)

Goal: Close all remaining live test orchestration gaps so that any disease can flow through Stages 1-3 without manual intervention. This stabilizes the foundation for the harder structural work.

Phase 2B — STRING Interaction Population (closes Gap 6, 8, 9 from live test) - New function: `populate_string_interactions(net)` reads `~/INTERCEPTA/data/string/9606.protein.links.v12.0.txt.gz` (98MB) and `9606.protein.aliases.v12.0.txt.gz` for ID mapping - Filter: `combined_score ≥ 700` (STRING high-confidence threshold) - Verification: GBM net should have EGFR↔MET edge after enrichment (canonical NSCLC/GBM bypass biology, currently absent per Finding 9) - Verification: At least 10 known cancer-relevant edges (EGFR↔ERBB2, EGFR↔HRAS, PTEN↔PIK3CA, TP53↔MDM2, BRAF↔MAP2K1, etc.) should appear - Adds to `intercepta_pipeline_v0.py` as new function; preserves existing functions

Phase 2C — AlphaFold Structure Joining (closes Gap 11 from live test) - New function: `attach_alphafold_structures(net)` queries AlphaFold DB for each protein in disease net - Cache structures in `~/INTERCEPTA/data/alphafold/cached/{uniprot_id}.pdb` - Verification: GBM net should have structures attached for EGFR, IDH1, TP53, PTEN, VEGFA — all top 5 GBM oncogenes - Output: enriched net with `genes[gene]['structure_path']` populated

Phase 2D — ChEMBL Coverage Expansion (closes Gap 7 from live test) - Replace local mCRPC-curated ChEMBL cache with broader query - Two implementation options to evaluate: (a) bulk ChEMBL download for all human protein targets, (b) live ChEMBL API queries cached per-disease - Verification: GBM net's EGFR should have hundreds of compounds (currently 0); BRAF should match its current 991 - Output: enriched net with `genes[gene]['compounds']` populated for any disease

Phase 2E — ClinicalTrials.gov Novelty Index (closes Gap 11/8 from live test, Vision Layer 9 partial) - Either: download CT.gov dump, build local searchable index - Or: live API integration (NIH CT.gov API v2) - Function: `check_novelty(smiles_or_drug_name)` returns “in trial / approved / never tested” - Verification: olaparib + abiraterone returns “approved combination, PROpel”; novel SMILES returns “not in trials”

Phase 2F — Pareto Generalization (closes mCRPC-only naming in `pareto_ranking.py`) - Refactor `build_mcrpc_candidates()` → `build_candidates(disease)` - Functions become disease-agnostic - Verification: works for GBM, AML, mCRPC with same call signature

Workstream A total scope: ~5 modular phases, each 2-4 days of focused work. Each adds one function to `intercepta_pipeline_v0.py` with verification assertions. Approximately 2-3 weeks if done sequentially.

Workstream B: Time Machine Generalization (Estimated 4-6 weeks, includes data acquisition)

Goal: RNA velocity working on at least one non-mCRPC cancer dataset, with per-cluster KAALCURA application demonstrating identification of pre-resistant subpopulations.

B1 — Acquire velocity-ready cancer dataset - Research: which cancer scRNA-seq datasets are publicly available with spliced/unspliced layers (loom, h5ad with velocity layers, or BAMs amenable to velocity) - Candidate datasets to evaluate (will research before committing): - Wouters et al. 2020 melanoma scRNA-seq with velocity - Becker et al. colorectal scRNA-seq if velocity-ready - GSE145281 or similar 10X cancer dataset with raw BAMs - Output: at least one cancer dataset on disk with confirmed spliced/unspliced layers - Time estimate: 1-2 days research + 0.5-3 days download (depends on dataset size, may exceed 5GB)

B2 — Adapt step3_run_scvelo.py to generic 10X velocity format - Current script expects STARsolo velocity_out/*_spliced.mtx format - Generalize to handle: loom files directly, h5ad with spliced/unspliced layers, BAMs + velocity integration - Output: scvelo runs on any velocity-ready input - Time: 2-3 days

B3 — Run scVelo on chosen dataset - Execute: full RNA velocity pipeline (filter_and_normalize, moments, recover_dynamics, velocity, velocity_graph, latent_time) - Compute: latent_time per cell, transition probability per cell, velocity confidence - Output: AnnData with velocity layers, latent_time per cell - Time: 1 day actual runtime (recover_dynamics is 30-60 min per scVelo docs) + 1-2 days troubleshooting

B4 — Identify pre-resistant cell populations - Cluster cells by Leiden - Apply RNA velocity to identify cells with high unspliced/spliced ratio for known resistance genes - Compare velocity-based “transitioning” cells to expression-based clusters - Output: per-cell “pre-resistance score” + cluster assignments - Time: 2-3 days

B5 — Apply KAALCURA per cluster - Compute R_prolif, R_emt, R_ddr per cluster (not bulk) - Identify which clusters are sensitive to which drug classes - Output: per-cluster KAALCURA axis scores + drug sensitivity predictions - Time: 1-2 days

B6 — Validate - For the chosen cancer, do the velocity-identified pre-resistant cells match known resistance biology? - If melanoma: do MITF-low/AXL-high cells appear as pre-resistant cluster? (Known melanoma resistance trajectory) - If colorectal: do EMT-high cells appear as pre-resistant? - Output: validation report - Time: 2-3 days

Workstream B total scope: ~6 phases, ~4-6 weeks. Largest open risk: data acquisition (B1) may be harder than estimated. If no clean velocity-ready cancer dataset exists, B1 expands to BAM acquisition + velocity run, which can take weeks for large datasets.

Workstream C: Universal ODE Refactor + Generative Chemistry (Estimated 6-8 weeks)

Goal: ODE accepts disease schema as input. First novel molecule produced by real generative chemistry tool.

C1 — ODE schema design - Define DiseaseSchema dataclass: cell_states (list), transition_rate_priors (dict), drug_pk_lookup (dict), validation_anchors (list of trial outcomes) - Refactor intercepta_three_mechanism_ode.py so all hardcoded constants (S_ARDEP, MU_S_TO_M, BRCA_FRAC) become schema attributes - Verification: existing mCRPC ODE results reproduce when called with mCRPC schema - Time: 1-2 weeks (careful refactor, must preserve Round 1 validation)

C2 — AML schema construction - Define AML schema: cell_states = ['blasts_sensitive', 'LSC_resistant', 'differentiated'], transition rates from BeatAML literature, drug PK from DrugBank for AML standard drugs (cytarabine, daunorubicin, venetoclax, gilteritinib) - Run ODE on AML schema, compare to BeatAML clinical outcomes - Document where it works and doesn't - Time: 2-3 weeks (literature curation is the bottleneck)

C3 — REINVENT4 install on Apple Silicon - Set up separate conda env (Python 3.10 per REINVENT4 reqs) - Install REINVENT4 from lockfile - Validate with simple smoke test (generate 100 SMILES with default config) - Time: 0.5-3 days (depends on dependency resolution, Apple Silicon ARM64 wheel availability — real risk per macOS partial support)

C4 — Target preparation - Choose one validated GBM target with strong vulnerability evidence: EGFR (top GBM gene, score 0.65, AlphaFold + PDB structures available) - Prepare target: AlphaFold structure, binding site definition, ATP-binding pocket coordinates - Generate REINVENT4 config: scoring components for binding (Vina docking), drug-likeness (QED), synthesizability (SAscore or similar) - Time: 3-5 days

C5 — Generation run - REINVENT4 generates 1000-10000 SMILES against EGFR ATP pocket - Filter: novel relative to ChEMBL EGFR inhibitors - Filter: drug-like (Lipinski + QED > 0.5) - Filter: synthesizable (SAscore < 5) - Output: top 50 novel candidate SMILES - Time: 2-5 days (depending on generation iterations and CPU speed on Apple Silicon)

C6 — Computational validation - AutoDock Vina docking of all 50 candidates against EGFR - ADMET-AI predictions for all 50 - Selectivity: dock against panel of related kinases (HER2, JAK2, etc.) to estimate off-target - Output: ranked list of novel EGFR candidates with full computational profile - Time: 3-5 days

C7 — Honest reporting - Document: novel SMILES list, docking scores, ADMET predictions, selectivity estimates - Honest about limitations: no wet-lab validation, computational only, BBB penetration not addressed for GBM - Output: first INTERCEPTA-generated novel molecule artifact - Time: 2-3 days

Workstream C total scope: ~7 phases, 6-8 weeks. Largest risks: C3 (REINVENT4 Apple Silicon install) and C2 (AML literature curation depth).

5.3 Horizon 1 Sequencing

The three workstreams interleave. A reasonable cadence over 12 weeks:

Weeks 1-3: Workstream A (Pipeline Orchestration) Focus on closing remaining live test gaps. Each phase 2-4 days. Builds momentum and stabilizes the pipeline before harder structural work.

Weeks 4-7: Workstream B (Time Machine) starts in parallel with C1 (ODE schema design) B1 data acquisition can happen in background while ODE refactor starts. Both are independent.

Weeks 8-9: Workstream B finishes, Workstream C continues Time Machine validated for one cancer. ODE refactor done. AML schema construction begins.

Weeks 10-12: Workstream C finishes REINVENT4 installed, target prepared, generation run, first novel molecule reported.

This is realistic but tight. Slippage of 2-4 weeks is plausible given dependency risks (REINVENT4 install, dataset acquisition, AML literature curation).

5.4 Horizon 1 Decision Points (Where Prasad and Claude Combined Decide)

Specific points where work pauses for explicit decision:

Decision Point 1 (after Phase 2B-2F): Are we ready for structural work, or are there orchestration gaps still surfacing in daily use?

Decision Point 2 (after B1 dataset selection): Which cancer dataset? This determines the Time Machine validation biology.

Decision Point 3 (after B6 validation): Did velocity-based pre-resistance match known biology? If not, what does that tell us about the Time Machine claim?

Decision Point 4 (after C1 ODE refactor): Does the refactored ODE reproduce Round 1 mCRPC results exactly? If not, the refactor introduced regressions.

Decision Point 5 (before C5 generation run): Confirm EGFR is the right first target. If we should test multiple targets, plan that.

Decision Point 6 (after C7 first novel molecule): What is the next disease (Round 3 NSCLC) for the Time Machine + ODE + Generative Chemistry stack?

5.5 Horizon 1 What's Excluded (and Why)

Explicit non-goals for the 90-day plan:

- **Phase 5 first pharma deliverable** — Prerequisites (universal pipeline + first novel molecule + ADMET integration) are all in Horizon 1 but the deliverable itself comes after
- **NSCLC, PDAC, Alzheimer's expansion** — Round 3+ is Horizon 2
- **Vision Part 6 diagnostic/predictive layer** — entire chapter unbuilt; not a 90-day scope
- **Universal net layers 11-14 (microbiome, epigenome, anatomy, pathogens)** — Net Spec Phases C-D, multi-month each
- **Self-improving feedback loop** — requires clinical partnerships (Open Question 4.6), years to build

- **Wet-lab partnership** — Open Question 4.1, not within current resources
- **Funding pursuit** — Open Question 4.2, separable from technical work
- **Publication** — first paper after Workstream C produces real artifact

5.6 Horizon 1 Risk Register

Risks ordered by probability × impact:

HIGH RISK: Workstream B data acquisition (B1) - Probability: 60% that finding velocity-ready non-mCRPC cancer data takes longer than 1 week - Impact: Time Machine blocked - Mitigation: Identify backup approach using BAM + velocityto (longer but always feasible)

HIGH RISK: REINVENT4 Apple Silicon install (C3) - Probability: 40% of installation friction lasting >3 days - Impact: Generative chemistry blocked; Workstream C can't complete - Mitigation: Have alternative tool (DiffSBDD, MolDiffusion) as backup

MEDIUM RISK: ODE refactor regression (C1) - Probability: 30% that refactor breaks Round 1 validation reproduction - Impact: Round 1 status becomes uncertain; rebuild required - Mitigation: Comprehensive test suite before refactor; abort and rollback if regression detected

MEDIUM RISK: Time Machine pre-resistance validation fails (B6) - Probability: 30% that velocity-identified clusters don't match known resistance biology - Impact: Time Machine claim weakened or refuted - Mitigation: This is itself information. Failure to validate means revise the vision claim, not fake the result.

LOW RISK: Workstream A orchestration - Probability: 10% of any single phase blocking >1 week - Impact: Minor — orchestration is well-understood code - Mitigation: Each phase is a function, not architecture. Easy to retry.

5.7 Horizon 1 Output Artifacts

When Horizon 1 is complete, the following artifacts exist:

1. `intercepta_pipeline_v0.py` extended to ~800-1200 lines, with 10+ working functions for any disease
2. New `intercepta_disease_schema.py` defining the schema architecture for ODE
3. `intercepta_universal_ode.py` accepting schemas, validated for mCRPC + AML
4. `intercepta_time_machine.py` running scVelo on any 10X velocity-ready dataset, producing per-cluster KAALCURA scores
5. `intercepta_generative_chemistry.py` wrapping REINVENT4 with INTERCEPTA-aware scoring
6. `~/INTERCEPTA/results/horizon1_first_novel_molecule.json` — actual SMILES + docking + ADMET + selectivity for first INTERCEPTA-discovered candidate
7. `~/INTERCEPTA/results/horizon1_time_machine_{cancer}.h5ad` — velocity-processed AnnData

8. ~/INTERCEPTA/results/horizon1_aml_ode_validation.json — universal ODE applied to AML

Plus updated documentation: closure docs for each workstream, updated risk register, updated alignment map showing what moved from MAJOR-DISTANCE / UNBUILT to PARTIAL or ALIGNED.

5.8 Horizon 1 Resource Implications

Compute: Apple Silicon MacBook Air sufficient for most work. Generative chemistry generation may benefit from cloud GPU but CPU runs are feasible for 1000-10000 molecule generation runs. Estimated cloud cost if needed: \$100-\$500.

Software: All planned tools are free / open-source (REINVENT4 Apache 2.0, scVelo, AutoDock Vina, ADMET-AI). RDKit installation needed (free).

Time: ~12 weeks of co-founder time (Prasad + Claude session work). Parallelism between workstreams reduces calendar time.

External help needed: None for Horizon 1 itself. Wet-lab and regulatory engagement deferred to Horizon 2+.

Horizon 1 is the most concrete piece of this plan because we have evidence to ground every claim. Horizon 2 (Sections 6) is more directional. Horizon 3 (Section 7) is strategic. # Section 6 — Horizon 2: Mid-Term Directional Plan (3-12 Months Past Horizon 1)

Less detail because dependencies between phases reveal themselves only as earlier phases complete. But the structure is concrete enough to plan.

6.1 Horizon 2 Strategic Frame

The mid-term question: Once Horizon 1's three structural gaps are closed, what does it take to actually produce a complete pharma deliverable for one disease and prove the universal claim across multiple diseases?

The mid-term answer: Phase 5 of vision — first pharma deliverable for mCRPC, then expand to NSCLC and AML in parallel, then publish.

Mid-term success criteria: 1. **Complete Phase 5 first pharma deliverable produced** — All 10 sections of Vision Part 9.1 populated with real content for one INTERCEPTA-discovered candidate 2. **Universal pipeline validated across ≥ 3 diseases** — mCRPC + AML + one of NSCLC/PDAC running end-to-end 3. **First peer-reviewed publication submitted** — Vision Part 11 Phase 5 says “publication: first peer-reviewed paper demonstrating novel drug candidate discovery from first principles” 4. **Self-improving loop demonstrated** — Round 3 (NSCLC) net-build time substantially shorter than Round 1 because of accumulated infrastructure

6.2 Horizon 2 Workstreams

Workstream D: Phase 5 First Pharma Deliverable Assembly (Months 1-3 of Horizon 2)

After Horizon 1 produces a first novel molecule (Workstream C output), assemble the complete 10-section deliverable for it.

The 10 sections per Vision Part 9.1, mapped to Horizon 1 outputs:

Section	Source from Horizon 1
1 Molecular structure	Workstream C7 output (novel SMILES + 3D from generative + Vina docking pose)
2 Mechanism of action	From disease net traversal (Phase 2B-2D outputs) — explain which net nodes the molecule targets, why this kills disease cells
3 Predicted clinical outcomes	From universal ODE (Workstream C1-C2) — apply the candidate to disease schema, simulate virtual cohort, produce HR estimate with documented confidence intervals
4 Resistance profile	From Time Machine (Workstream B) — does the candidate target the pre-resistant population identified by velocity?
5 Combination rationale	From Scout 3 + ODE — if the candidate is part of a combination, why does the combination kill both populations?
6 Safety profile	From ADMET-AI integration (closing Vision Stage 4 Layer E)
7 Synthesis route	From ASKCOS retrosynthesis integration (closing Layer F universally; currently mCRPC-only)
8 Novelty confirmation	From CT.gov novelty index (Phase 2E)
9 vs SoC comparison	From universal ODE simulating SoC vs candidate side-by-side
10 Suggested trial design	New module — biomarker selection from disease net, primary endpoint from ODE simulation, dose range from PK/PD

Decision point at start of Workstream D: Which disease for first deliverable? Three reasonable options:

- **Option a: mCRPC (PROpel-style novel addition)** — best validation ground truth, but combination partners are existing approved drugs
- **Option b: GBM** — first INTERCEPTA disease that's not mCRPC, biggest novelty value if we produce a credible novel BBB-penetrant EGFR inhibitor
- **Option c: AML** — Round 2 net skeleton already validated; Workstream C produces novel for AML target if we choose AML for C generation run

Decision deferred to Horizon 1 closure point.

Workstream E: Round 3 — NSCLC End-to-End (Months 2-6 of Horizon 2)

NSCLC is Vision Round 3. Largest cancer burden, well-characterized EGFR/KRAS/ALK biology, multiple immunotherapy trials for validation.

E1: Build NSCLC disease net — using completed pipeline orchestrator from Horizon 1 Workstream A. This should be a 1-day function call now, not a months-long custom build.

E2: Acquire NSCLC scRNA-seq — for Time Machine application. Real datasets exist (Zilionis et al. 2019, Lambrechts et al. 2018, Maynard et al. 2020 EGFR-resistance dataset).

E3: Build NSCLC ODE schema — cell states (lung adenocarcinoma vs squamous, EGFR-mutant vs WT, immune microenvironment populations), transition rates from literature, drug PK for osimertinib, pembrolizumab, etc.

E4: Validate against NSCLC trials — KEYNOTE-024 (pembrolizumab vs chemo), FLAURA (osimertinib vs gefitinib/erlotinib), CodeBreaK 200 (sotorasib for KRAS-G12C). These are the validation anchors per Vision Part 7.2.

E5: Generate novel candidates for NSCLC targets — using completed generative chemistry from Horizon 1 Workstream C. Targets: EGFRvIII, KRAS-G12D, ALK fusion variants.

E6: Round 3 closure with documented limitations

Calendar: ~4 months of dedicated work for NSCLC end-to-end if Horizon 1 infrastructure is solid.

Workstream F: Self-Improving Loop Foundations (Months 4-9 of Horizon 2)

Vision Part 7.3 + 12.1 — the system gets faster with each disease.

F1: Cross-disease molecular transfer — when Workstream E produces NSCLC candidates, automatically check if any work for mCRPC or AML based on shared net nodes (e.g., KRAS-G12D candidates might be relevant to PDAC)

F2: Net completeness tracking — each disease processed adds STRING/ChEMBL/AlphaFold data populations. Document net coverage growth: how many genes have full layer population after Round 1, Round 2, Round 3?

F3: Validation parameter learning — when Round 3 NSCLC validation shows ODE has different bias than Round 1 mCRPC, the schema captures this. Learn parameters per disease.

F4: Literature mining integration — vision Net Spec Step 18, automated PubMed updates to disease nets. Real-time integration of new findings.

Calendar: Ongoing throughout Horizon 2. F4 is a heavier project (could be Horizon 3).

Workstream G: First Publication (Months 6-9 of Horizon 2)

Vision Phase 5 deliverable: peer-reviewed paper.

G1: Choose target journal — Nature Medicine? Cell? eLife? Different audiences, different review timelines. Computational drug discovery papers often appear in Nature Communications, J. Cheminformatics, Bioinformatics.

G2: Define paper scope — likely “INTERCEPTA: a universal computational platform for novel drug discovery, validated against PROpel and FLAURA” or similar. Demonstrates: (a) validation against trials, (b) novel candidate generation, (c) wet-lab-relevant claims with explicit limitations.

G3: Submit + revise — typical Nature-tier review is 6-12 months from submission.

Calendar: Writing in months 6-7, submission month 8, first decision month 10-12.

6.3 Horizon 2 Decision Points

Mid-Horizon Decision Point 1 (Month 3): Did Phase 5 first deliverable produce credible output? If yes, Workstream E (NSCLC) gets full priority. If no, what gap blocked it and how to address?

Mid-Horizon Decision Point 2 (Month 6): NSCLC validation — does universal ODE applied to NSCLC produce validation results comparable to mCRPC? If significantly worse, the universal claim has a problem.

Mid-Horizon Decision Point 3 (Month 9): Publication readiness — do we have publishable results? If yes, submit. If no, identify what’s missing and decide whether to extend Horizon 2 or revise scope.

6.4 Horizon 2 Risk Register

MEDIUM RISK: Workstream D scope creep — assembling 10-section deliverable may surface that some sections require months of additional infrastructure work. Mitigation: explicit per-section gap analysis in first 2 weeks of D.

MEDIUM RISK: Horizon 1 infrastructure not robust enough — running NSCLC may surface bugs that GBM didn’t. Mitigation: same approach as today’s live test methodology.

MEDIUM RISK: Publication review cycles slow — typical ML/computational biology papers see 4-12 months review. Mitigation: parallelize publication writing with technical work; don’t gate next-disease work on previous publication.

LOW-MEDIUM RISK: Wet-lab validation pressure — at some point in Horizon 2, “where’s your wet-lab data?” becomes a recurring question from any external reviewer. Mitigation: establish wet-lab partnership search as Horizon 2 priority (see Section 7 Horizon 3).

OPEN RISK: Funding — if Prasad's resources don't cover ~12 months of focused work, this horizon may not complete. Plan honestly does not assume external funding.

6.5 Horizon 2 Resource Implications

Compute: Likely needs cloud GPU at some point (full pharmacome screening, larger generative runs). Estimated \$1K-\$5K cloud spend over 12 months.

Software: Same free tools as Horizon 1. May need DrugBank academic license (~\$0 academic) or commercial license if not academic.

Time: ~12 months of focused co-founder work. Calendar slip 6 months would not be surprising.

External help: First wet-lab inquiry and first publication submission both require external touchpoints. Plan needs: - Wet-lab partner identification (academic preferred for cost) - Journal selection + review cycle awareness - Possibly: first conference presentation as visibility step

Funding: If Horizon 2 produces credible publication-bound results, this is the natural inflection point for fundraising or partnership conversations. Plan does not assume but does not preclude.

6.6 Horizon 2 Output Artifacts

When Horizon 2 closes, the following artifacts exist:

1. Phase 5 first pharma deliverable — complete 10-section package for one INTERCEPTA candidate (Workstream D)
2. NSCLC universal pipeline output — net, candidates, validation against trials (Workstream E)
3. AML extended through novel candidate generation (carryover Workstream C/E continuation)
4. Cross-disease molecular transfer evidence (Workstream F1)
5. First peer-reviewed publication submission (Workstream G)
6. Updated alignment map: many MAJOR-DISTANCE entries should now be PARTIAL or ALIGNED
7. Updated open questions: some questions answered through experience, new ones surfaced

6.7 Horizon 2 Strategic Outputs (Not Code)

Beyond technical artifacts, Horizon 2 produces:

- **Wet-lab partnership conversations initiated** — even if no wet-lab agreement signed, the conversation pipeline started
- **Publication-driven academic visibility** — peer review process surfaces which vision claims hold up to expert scrutiny

- **Pharma awareness** — first deliverable, even if not commercially licensed, becomes the artifact pharma can be shown
- **Regulatory familiarization** — documentation in deliverable format starts shaping toward regulatory submission requirements (Vision Part 12.3)

These soft outputs are as important as the code artifacts. They turn INTERCEPTA from “private development project” into “real platform with external validation pathway.” # Section 7 — Horizon 3: Long-Term Strategic Plan (Year 2-5+ to Full Vision Success)

This horizon is directional, not detailed. The honest reason: dependencies between phases two years out cannot be planned with the same rigor as 90 days. What I can do is define the strategic structure: what unlocks what, what the major decision points are, what success looks like at scale.

7.1 Horizon 3 Strategic Frame

The full-vision question: What does it take for INTERCEPTA to become the universal computational drug discovery platform the vision describes?

The full-vision answer: Six things must converge:

1. **Technical universality** — pipeline runs end-to-end for diverse diseases (cancer + neurodegenerative + infectious + rare)
2. **Self-improvement** — measurably faster per disease as the system grows
3. **Wet-lab validation pathway** — INTERCEPTA candidates routinely confirmed experimentally
4. **Pharma adoption** — at least one pharma partner using INTERCEPTA outputs as primary R&D input
5. **Regulatory acceptance** — at least one INTERCEPTA candidate enters clinical trial
6. **Mission demonstration** — INTERCEPTA-discovered drug saves lives (the ultimate metric)

These have wildly different timelines. Items 1-2 are 2-3 year. Items 3-4 are 3-5 year. Items 5-6 are 5-10+ year.

The Horizon 3 strategic question is not “what code do we write” but “what milestones do we hit, in what order, to make these six things real.”

7.2 Horizon 3 Major Phases

Phase H3-A: Technical Universality (Year 1-2 of Horizon 3, i.e. Year 2-3 from now)

By end of Horizon 2, INTERCEPTA runs for mCRPC + AML + NSCLC. Phase H3-A extends this to:

- **Round 4 PDAC** — pancreatic adenocarcinoma, near-zero current treatments, 90% KRAS mutations, dense stromal microenvironment

- **Round 5 Alzheimer's** — first non-cancer disease, BBB penetration constraint, multi-target architecture explicitly designed for this
- **Round 6 Drug-resistant TB** — first pathogen disease, bacterial biology not human cell biology, novel antibiotic generation

Each round validates that the universal pipeline + universal ODE + Time Machine + Generative Chemistry actually works across diverse disease classes. The point is not just to add diseases but to surface where the universal claim breaks for specific biology.

Open scientific question: Does the two-population ODE meaningfully apply to non-cancer? Alzheimer's has different dynamics. TB has bacterial replication. This is when the "universal" claim is genuinely tested.

Output: Six diseases with end-to-end pipeline runs. Documentation of where the architecture extends and where it requires extension.

Phase H3-B: Wet-Lab Validation Pathway (Year 1-3 of Horizon 3)

This is the open question 4.1 made concrete. Horizon 3 must address it because computational candidates without wet-lab data have limited external value.

Strategy options:

Strategy 1: Academic partnership - Identify 1-2 academic labs willing to validate INTERCEPTA candidates experimentally - Co-publication model per Vision Part 12.4 — they get first authorship on validation papers, INTERCEPTA gets validation - Cost: low (academic labs absorb cost in their own grants) - Timeline: 6-12 months to identify partner, then ongoing - Requirement: Prasad's network or new outreach

Strategy 2: CRO contract - Pay a contract research organization (CRO) to validate top candidates - Cost: high (\$50K-\$500K per candidate depending on assay depth) - Timeline: faster (months not years) - Requirement: funding

Strategy 3: Direct pharma partnership where pharma validates - Pharma takes INTERCEPTA candidate, runs internal preclinical validation - Cost: pharma absorbs - Timeline: pharma-dependent, often slow - Requirement: pharma interest, which requires (Phase 5 deliverable from Horizon 2)

Strategy 4: Build internal wet-lab - Set up basic biology validation capability - Cost: very high (\$1M+ initial, ongoing operations) - Timeline: 1-2 years to set up - Requirement: significant funding, hiring, space

For Horizon 3 planning purposes, assume **Strategy 1 (academic partnership)** as primary, with **Strategy 3 (pharma-validated)** as secondary if Horizon 2 deliverable produces interest.

Phase H3-C: Pharma Adoption (Year 2-4 of Horizon 3)

Vision Part 9.2 business model: pharma pays \$300K-\$500K per disease for complete drug candidate package. Horizon 3 must move from “we produced one deliverable” (end of Horizon 2) to “pharma partners are using INTERCEPTA outputs.”

Adoption pathway:

Step 1: Demonstration deliverables (early Horizon 3) - Publish complete deliverables for mCRPC + AML + NSCLC openly - These become reference artifacts pharma can evaluate

Step 2: First paid engagement - Pharma BD conversations driven by published demonstration - First contract: likely a “validation engagement” — pharma asks INTERCEPTA to find candidates for a target they’re already working on, compares to their internal results - Cost: <full deliverable price; relationship-building

Step 3: Standard contract - Once validation engagement succeeds, pharma engages on diseases they’re not already working on - Full deliverable price (\$300K-\$500K per disease per Vision) - Multiple parallel contracts as the model proves out

Decision point: Open vs proprietary. Vision Part 8.2 says discovery method is open, molecules can be patented. In practice pharma may want exclusivity for their deliverables. The plan needs to navigate: what stays open, what’s customer-confidential, what’s IP that goes to pharma.

Phase H3-D: Regulatory Pathway (Year 2-5 of Horizon 3)

Vision Part 12.3 — regulatory awareness from Day 1. Horizon 3 makes this real.

First IND-enabling work: Will be done by pharma partner after they license a candidate. INTERCEPTA’s role: provide regulatory-ready documentation in the deliverable.

INTERCEPTA’s regulatory deliverable contributions: - ADMET predictions formatted to match IND submission requirements (Vision Part 12.3) - Companion diagnostic biomarker identification (from disease net) - Regulatory pathway analysis (standard / fast track / breakthrough therapy / RMAT) — requires regulatory expertise we don’t have, will need consultant - Manufacturing (CMC) — entirely pharma’s responsibility

External help needed for Horizon 3-D: A regulatory affairs consultant. Cost: \$50K-\$200K per engagement. Timing: when first INTERCEPTA candidate approaches IND.

Phase H3-E: Self-Improving Loop Realization (Year 1-3 of Horizon 3)

Vision Part 7.3 + 12.1. The system measurably gets faster per disease.

Concrete metrics for self-improvement:

- **Net-build time:** Round 1 mCRPC took months. Round 2 AML net skeleton took weeks. Round 3 NSCLC should take days. Round 5+ should be hours-to-days.
- **Validation hurdle:** For each new disease, document validation AUROC/HR achievement. Self-improvement = the bar is met faster, not lowered.

- **Cross-disease transfer hits:** What % of candidates for new disease come from prior disease screens? Increasing % = self-improving working.
- **Literature mining freshness:** Time from new published finding to integration into disease net. Vision spec target: continuous (real-time integration).

Phase H3-E technical deliverables:

- F4 from Horizon 2 (literature mining) extended to real-time
- Federated learning architecture for clinical outcome integration (Vision Net Spec Part 7 — novel tech to develop)
- Cross-disease transfer learning (Net Spec Step 20)
- Net coverage tracking dashboard

Phase H3-F: Diagnostic and Predictive Layer (Year 2-5 of Horizon 3)

Vision Part 6 — entire chapter currently UNBUILT per alignment map.

This is a MAJOR scope expansion. Goes from drug discovery platform to disease management platform. Includes:

- **6.1 Current disease identification module** — patient sample → disease net signature match → diagnosis
- **6.1 Treatment response prediction module** — patient profile + INTERCEPTA disease net → personalized drug ranking
- **6.1 Resistance monitoring module** — longitudinal sample tracking, pre-resistance fraction trending
- **6.2 Polygenic risk + transcriptional trajectory predictor** — future disease risk
- **6.2 Population surveillance** — emerging disease signals from aggregated data
- **6.3 Future disease modeling** — pathogen evolution, synthetic biology threat assessment

Each of these is its own multi-year project. Horizon 3 either: (a) starts these as concurrent workstreams, (b) defers all of Part 6 to Horizon 4, (c) selects the highest-value subset to build.

Recommendation: Horizon 3 builds 6.1.3 (treatment response prediction per patient) as an extension of the drug discovery platform. The other Part 6 capabilities defer.

Phase H3-G: Universal Net 15 Layers Completion (Year 1-4 of Horizon 3)

Net Spec 20-step plan. Horizon 1 + 2 cover Phases A + B partially. Horizon 3 completes:

- **Phase C (Universal Expansion):** Steps 8-13 — DisGeNET full, HMDB full integration, AlphaFold all targets, Human Cell Atlas integration, ENCODE/Roadmap epigenome, DICE/CIBERSORTx immune full
- **Phase D (Future-Proofing):** Steps 14-17 — pathogen genomes, microbiome, spatial transcriptomics, full ZINC pharmacome
- **Phase E (Self-Improving):** Steps 18-20 — already covered in Phase H3-E above

These steps require sustained engineering. Each layer is weeks-to-months. Phase H3-G is genuinely a 2-3 year program.

7.3 Horizon 3 Strategic Decision Points

Year 2 Strategic DP: Continue solo (Prasad + Claude) or scale team? Several Horizon 3 phases require capability we don't currently have (regulatory expertise, wet-lab management, software engineering at scale, BD/sales).

Year 2 Strategic DP: Open or commercial? Vision says open science. But Horizon 3 funding likely requires some commercial component. Decide which deliverables stay open, which become commercial.

Year 3 Strategic DP: Disease prioritization — at this point we have ~10 diseases working. Which disease gets candidate-to-clinical-trial priority? This determines which mission demonstration we pursue first.

Year 3-4 Strategic DP: Funding model — academic grants (NIH, NSF, foundations), private sector (VC, pharma), public-benefit corporation? Different choices have different obligations.

Year 4-5 Strategic DP: Mission demonstration — first INTERCEPTA-discovered drug entering Phase 1 trial. Which disease? Which candidate? Who runs the trial?

7.4 Horizon 3 Risk Register

EXISTENTIAL RISK: No funding — without funding past Horizon 2, Horizon 3 stalls. Plan must address.

EXISTENTIAL RISK: Vision claim fails fundamentally — if Time Machine doesn't generalize, or universal ODE doesn't work for non-cancer, or generative chemistry produces only synthetically-infeasible molecules, the universal claim breaks. Need vision-revision capability.

HIGH RISK: Wet-lab validation produces no positive results — if the first 5-10 INTERCEPTA candidates all fail wet-lab, the platform's credibility is questioned. Need to plan for low-positive-rate reality.

HIGH RISK: Pharma adoption slow — pharma is conservative. INTERCEPTA may need 3-5 years before first paid contract.

MEDIUM RISK: Team scaling failure — even with funding, hiring scientific co-founders / senior contributors is hard. Plan for sustained two-person execution as default.

MEDIUM RISK: Regulatory rejection of computational evidence — FDA may not accept computational predictions as substantial evidence. Plan for wet-lab as primary, computational as supplementary.

MEDIUM RISK: Patent landscape conflicts — INTERCEPTA-generated novel molecules may inadvertently infringe existing patents (which are not in ChEMBL). Need IP search capability before claiming novelty.

7.5 Horizon 3 Success Definition

After 5 years from today (May 2031), full vision success looks like:

1. INTERCEPTA pipeline running in production for ≥ 10 diseases (cancer, neurodegenerative, infectious, rare)
2. Self-improving loop demonstrably reducing per-disease time (e.g., < 1 month for new disease)
3. ≥ 3 INTERCEPTA-discovered candidates in active wet-lab validation at academic or pharma partners
4. ≥ 1 INTERCEPTA-discovered candidate in IND-enabling preclinical studies
5. ≥ 3 pharma partnerships actively engaging INTERCEPTA for drug discovery
6. ≥ 3 peer-reviewed publications validated by community
7. Open-published disease nets for ≥ 5 diseases used by other research groups
8. Funding model sustainable (whether grant, commercial, or hybrid)

This is realistic-ambitious for 5 years if Horizon 1 + 2 hit their marks.

Mission success (life-saving impact) is 7-10+ years from today. Plan does not over-promise on this. The drug discovery $\rightarrow$ trial $\rightarrow$ approval $\rightarrow$ adoption timeline is what it is.

7.6 Horizon 3 Course-Correction Mechanism

The plan acknowledges that Horizon 3 will be revised many times. Build the revision into the structure:

- **Quarterly: vision-revision checkpoint.** Re-read vision document. What claims still hold? What claims need updating? Update the alignment map.
- **Annual: strategic review.** All open questions revisited. Funding, team, partnership, regulatory status evaluated.
- **Per-round: validation review.** When each new disease completes its end-to-end run, document where the universal claim held and where it didn't. Vision document gets revised based on accumulated evidence.

The vision is the founding document. It is not immutable. It can and should evolve based on what reality teaches us.

This concludes the three-horizon plan. The remaining sections cover commitments and decision points where Prasad and Claude must combine to choose direction. # Section 8 — Principles and Decision Points

8.1 Operating Principles (Non-Compromise Commitments)

These are the principles you defined explicitly in our session. They govern all work below.

P1 — No artificial constraints filtering vision-right work. “Reasonable size under 5GB to fit tonight” is the kind of constraint that should never override vision-aligned work. Sessions serve the work; the work doesn’t fit sessions.

P2 — Honest validation, no manipulation. Round 1 closed with documented 2× systematic bias. Round 2 closed with KAALCURA plateau at 0.53. These are real, recorded, not hidden. Future rounds follow the same standard.

P3 — Research before code. Every micro-step of the vision is preceded by careful, strict, deep research. Including the present plan.

P4 — Fix structure, don’t tune. Three rounds at 0.53 AUROC for AML KAALCURA showed that tuning thresholds doesn’t fix structural problems. Structural gaps require structural fixes.

P5 — Preserve past work. No modifications to Round 1 / Round 2 artifacts (`disease_net_builder.py`, `intercepta_unified_ode_v4_1.py`, etc.) without explicit reason and rollback plan. New functionality is additive in new modules.

P6 — Find or develop novel technology when the vision requires it. Vision Net Spec Part 7 explicitly anticipates this. Multi-scale GNNs, temporal knowledge graphs, causal inference on biological graphs, federated learning, generative chemistry constrained by knowledge graph — these may need to be invented, not just integrated.

P7 — Combined CSO calls. Just Prasad + Claude. Decisions made together with both reasonings explicit. Claude makes recommendations as CSO; Prasad has final approval. No “delegate to Claude alone” pattern, no “wait for Prasad to decide every micro-thing” pattern.

P8 — Every minute precious because lives depend. This means use every minute on work that genuinely advances the vision, not work that feels productive while sidestepping hard problems.

P9 — Vision is updatable based on evidence. The vision document is the founding document. It is not the contract. As Horizons 1-3 produce evidence, the vision can and should evolve.

P10 — Truthful claims only. Never claim ALIGNED when reality is PARTIAL. Never claim PARTIAL when reality is UNBUILT. The alignment map in Section 3 is the standard.

8.2 Validation Gates

Before each phase advances to the next, explicit validation gates that must pass:

Horizon 1 Validation Gates

Gate 1A (Workstream A complete): All 19 live test gaps either closed or explicitly deferred with documented reason. Pipeline runs end-to-end for at least one new disease (NSCLC) without manual intervention.

Gate 1B (Workstream B complete): Time Machine runs on at least one cancer NOT mCRPC with biology-validated pre-resistant cluster identification. If validation against known biology fails, document and revise vision claim.

Gate 1C (Workstream C complete): First novel SMILES generated by real generative tool, docked, ADMET-evaluated, novelty-confirmed. Output is genuine new chemical entity not in ChEMBL.

Horizon 1 Aggregate Gate: All three structural gaps either closed or explicitly demonstrated as harder than projected with documented limitation.

Horizon 2 Validation Gates

Gate 2A: Phase 5 first pharma deliverable complete — all 10 sections populated with real content.

Gate 2B: Three diseases (mCRPC, AML, NSCLC) running end-to-end with comparable validation rigor.

Gate 2C: First peer-reviewed publication submitted.

Horizon 3 Validation Gates

Gate 3A (Year 2): Six diseases running. Universal claim tested across cancer + non-cancer.

Gate 3B (Year 3): Wet-lab validation pathway operational — at least one academic partner producing experimental data on INTERCEPTA candidates.

Gate 3C (Year 4): First IND-enabling work begun for at least one INTERCEPTA candidate.

Gate 3D (Year 5): First INTERCEPTA candidate in clinical trial.

8.3 Decision Points That Need Combined Resolution Before Horizon 1 Starts

Twelve specific decisions that I cannot make alone (or shouldn't, per P7). Each needs Prasad's input:

DP1 — Workstream sequencing

The plan proposes $A \rightarrow B+C$ parallel, but A could be pursued more or less aggressively before B and C start. Should A complete fully first (3 weeks pure orchestration), or interleave (orchestration in weeks 1-3, Time Machine starts week 4)?

DP2 — Time Machine cancer choice (Workstream B1)

Once velocity-ready cancer datasets are surveyed, which one? Implications for which biology we test first.

DP3 — Universal ODE first non-mCRPC application (Workstream C2)

AML is natural since net skeleton exists. NSCLC is more impactful. PDAC is hardest test case. Which disease for the universal-ODE first proof?

DP4 — Generative chemistry target (Workstream C4)

Plan proposes EGFR for GBM. Alternative targets: KRAS-G12D (NSCLC, bigger pharma interest), BCL-2 (AML, well-validated), IDH1-R132H (mutant-specific, novel space). Which one?

DP5 — REINVENT4 vs alternative

If REINVENT4 install on Apple Silicon fails, use which alternative? DiffSBDD, MolDiffusion, REINVENT3 (older but lighter), or build minimal generator from scratch?

DP6 — Cloud spend authorization

Some Horizon 1 work may benefit from cloud GPU. Estimated \$100-\$500. Authorize / decline / set spending limit?

DP7 — Wet-lab partnership timing

Horizon 2 needs wet-lab partner. When does outreach begin? Now (during Horizon 1) so partnership ready when Horizon 2 needs it? Or later?

DP8 — First publication scope

Horizon 2 produces publication. Scope: (a) full INTERCEPTA architecture paper, (b) specific result paper (e.g., novel EGFR inhibitor), (c) methods paper (e.g., universal ODE refactor). Different scopes target different audiences.

DP9 — IP strategy timing

At what point do we engage patent counsel? When first novel molecule produced? When first wet-lab validation positive? Or later?

DP10 — Funding pursuit timing

Plan does not assume funding. But fundraising activities (advisor conversations, NIH grant applications, foundation outreach) are work that can happen in parallel. Timing?

DP11 — Team expansion criteria

Under what specific conditions do we hire / bring on additional people? What roles first? (Software engineer for infrastructure? Wet-lab manager? Regulatory consultant?)

DP12 — Vision revision cadence

Plan says quarterly vision-revision checkpoint. Adopt this? Different cadence? Or only at horizon transitions?

These twelve decisions don't all need answering today. But they should be answered within the first 1-2 weeks of Horizon 1 execution because they affect immediate work direction.

8.4 What I Need From You Before We Start Horizon 1 Execution

Three minimum-viable answers needed:

MIN1: Approval of the three-horizon framing as the right structure. Or revision request.

MIN2: First-week direction — start Workstream A (orchestration), or different sequencing?

MIN3: Authorization to begin technical work on this plan, vs more iteration on plan itself.

The other 12 decision points can resolve as work progresses. They don't gate starting.

8.5 What This Plan Is Not

Honest about limitations of this plan itself:

- **It's not a budget.** No dollar amounts beyond rough cloud-spend estimates.
- **It's not a hiring plan.** Two people execute it as written.
- **It's not a fundraising plan.** Raising money is a separable activity.
- **It's not a partnership plan.** Specific pharma names, specific academic labs not named.
- **It's not a publication plan.** Specific paper outlines not provided.
- **It's not a marketing plan.** No positioning beyond what's in the vision document.

These are real plans that may need to exist, but they are downstream of the technical plan. Built when we have the artifacts to discuss with funders, partners, journals, etc.

This is the complete plan. Section 9 follows with a summary and proposed first action. #
Section 9 — Summary and Proposed First Action

9.1 Plan Summary in One Page

Where we are: INTERCEPTA today is a real research project for mCRPC drug discovery with universal aspirations. Round 1 mCRPC ODE validated with documented 2× systematic bias. Round 2 AML KAALCURA on patients plateaus at 0.53 AUROC (cell-line-to-patient transfer gap). May 2 GBM live test produced 19 findings; Phase 1 + 2A this session closed 8 of them. The pipeline now has 6 working functions for any disease in OpenTargets via `intercepta_pipeline_v0.py` (416 lines). Real foundation. Small relative to vision.

Where the vision says we're going: Universal computational drug discovery platform. Discovers novel drug molecules for any disease. Diagnostic + predictive system. Self-improving. Complete digital model of human biology with 15-layer net, 3M nodes, 10-50M edges. First pharma deliverable in Phase 5 (Vision Weeks 37-52). Expansion through 7 disease rounds covering all major diseases + 7000+ rare diseases.

The gap: ~30 vision claims UNBUILT, ~15 MAJOR-DISTANCE, ~15 MID-DISTANCE, ~25 PARTIAL, ~5 ALIGNED. Three structural gaps dominate: (A) Generative chemistry not built, (B) RNA velocity Time Machine not generalized beyond a placeholder script, (C) Two-population ODE hardcoded prostate-only.

The plan: Three horizons.

- **Horizon 1 (90 days):** Three workstreams in parallel. (A) Pipeline orchestration closure. (B) Time Machine generalization to one non-mCRPC cancer. (C) Universal ODE refactor + first novel molecule via REINVENT4. Output: foundational structural gaps closed.
- **Horizon 2 (3-12 months past Horizon 1):** Phase 5 first pharma deliverable assembled. Round 3 NSCLC end-to-end. Self-improving loop foundations. First peer-reviewed publication submitted.
- **Horizon 3 (Year 2-5+):** Six diseases through end-to-end pipeline. Wet-lab validation pathway operational. Pharma adoption. First IND-enabling work. Diagnostic + predictive layer. Universal net 15 layers complete. Mission demonstration: first INTERCEPTA-discovered drug enters Phase 1 trial.

The principles: No artificial constraints. Honest validation. Research before code. Fix structure not threshold. Preserve past work. Find/develop novel tech when vision requires. Combined CSO calls. Every minute precious. Vision is updatable. Truthful claims only.

The known unknowns: 12 open questions the vision documents don't specify (wet-lab bridge, funding model, team structure, validation criteria, regulatory engagement, clinical outcome feedback, IP strategy, data licenses, compute scale, first deliverable choice, vision-revision triggers, success definitions). Plan acknowledges and works with each.

9.2 Why This Plan Is Honest

This plan is honest by construction:

- Every reality claim cited to specific files or findings
- Every vision claim explicitly rated against reality
- Every horizon has explicit non-goals listed
- Every workstream has explicit risk register
- Every horizon has explicit validation gates
- Open questions named, not glossed over
- Resource implications stated, not hidden

- “It’s not a budget / hiring plan / fundraising plan / partnership plan” stated explicitly

It is not optimistic and not pessimistic. It is what evidence supports.

9.3 The Single First Action

After Horizon 1 framing is approved, the very first technical action is:

Workstream A Phase 2B — STRING Interaction Population for Any Disease.

Why this first: - Closes Gap 6 from live test (canonical EGFR→MET escape route currently absent for GBM) - Closes Gap 8 (universal net interactions mCRPC-centric) - Enables Vision Stage 2 escape route identification claim (currently broken for any non-mCRPC disease) - Highest leverage among orchestration gaps because escape routes appear at multiple vision points (Stage 2.5, Scout 4 network perturbation, Part 6.1 resistance monitoring) - Specific scope: ~30-50 lines of code, large file processing (98MB STRING file), well-defined success criterion (EGFR→MET appears in GBM net after enrichment) - Estimated 2-4 days of work

Concrete first deliverable: append to `intercepta_pipeline_v0.py` a function `populate_string_interactions(net, min_score=700)` that reads `~/INTERCEPTA/data/string/9606.protein.links.v12.0.txt.gz` plus `9606.protein.aliases.v12.0.txt.gz` for ID mapping, populates each gene’s interactions field with high-confidence STRING edges. Verification: assert EGFR→MET edge appears in enriched GBM net, plus 10+ other canonical cancer edges.

After Phase 2B passes verification, proceed to 2C (AlphaFold), 2D (ChEMBL), 2E (CT.gov), 2F (Pareto generalization). Workstream A complete in 2-3 weeks.

9.4 What I Need From You

Three minimum-viable answers (repeated from Section 8.4):

MIN1: Approval of the three-horizon framing as the right structure (or revision).

MIN2: First-week direction — start Workstream A Phase 2B (STRING interactions) as proposed, or different sequencing?

MIN3: Authorization to begin technical work on this plan, vs more iteration on plan itself.

If MIN1 = yes, MIN2 = yes, MIN3 = yes, we begin Phase 2B.

If revisions needed, tell me what’s wrong with the plan and I revise.

If you need time to read the plan first, take the time. The plan is now written; it doesn’t expire.

9.5 Closing Acknowledgment

This plan is built on every earlier session's work — Round 1 mCRPC closure, Round 2 AML KAALCURA validation, the May 2 GBM live test, Phase 1 + 2A pipeline foundation. It's also built on this session's deeper re-read of vision and net spec.

Mistakes I made this session that the plan corrects: - I used "fit in tonight" as a constraint that violated P1. - I made CSO calls without your input that ended up wrong (Option II RNA velocity tonight when no velocity-ready data on disk). - I let momentum override systematic ordering when proposing Phase 2B before Phase 2A. - I treated documentation as avoidance when you clarified that real vision work is the goal, but then proposed novel-molecule tonight without sufficient research.

The plan above is built so those patterns don't repeat. Each horizon has explicit gates. Each workstream has explicit scope. Each decision is named. Each risk is documented.

You and Claude, co-founders, doing this honestly. No compromise. Every minute precious because lives depend.

Ready when you are.

End of Plan v1.